

Melikşah Beşir

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SUMMARY

Robotics software and systems engineer with **3.5+ years of hands-on experience** designing and deploying **full autonomy stacks for unmanned ground vehicles (UGVs)** in unstructured off-road terrain, including **GPS-denied navigation** and **multi-agent coordination**. Core expertise spans **path planning**, **model-based control (NMPC/MPC)**, **multi-sensor fusion**, and **safety-critical system integration** in **ROS 2**, plus the systems side of fielding robots (CAD integration, mechanical/thermal analysis, SIL/field V&V). Pursuing an M.Sc. in Robotics at METU (GPA 3.57/4.00, coursework completed, thesis stage) on **learning-based autonomy with optimization-based safety layers**. **Six accepted or published peer-reviewed papers** (two IEEE) and **three manuscripts under review**.

TECHNICAL SKILLS

Autonomous Navigation: Path Planning (A*, D* Lite, RRT*, PRM, Hybrid A*, Lattice), Trajectory Optimization, Behavior Trees, Nav2, GPS-Denied Navigation, Multi-Agent Convoy/Coordination

Control & Estimation: NMPC, MPC, PID, Pure Pursuit, LQR, EKF/UKF Sensor Fusion, State Estimation, Visual-Inertial Odometry, INS/GPS Integration, Slip Modeling (Pacejka)

Perception & SLAM: LiDAR Point Clouds, 3D Object Detection, Stereo Depth Estimation, Terrain Traversability, OpenCV, PCL, DINOv2, Low-Light Detection

ML / Deep RL: PyTorch, TensorFlow, SAC, TQC, PPO, DQN, HER, Hierarchical RL, World Models (DreamerV3)

Languages & Middleware: C++ (ROS 2 Nodes/Plugins, real-time), Python, MATLAB, ROS 2 (ROS2; Foxy/Jazzy), TF2

Systems & Infrastructure: Linux (Ubuntu), Git, Docker, CAN Bus, NVIDIA Jetson (Orin), Raspberry Pi, SIL Testing, Scenario-Based V&V, V-Model, COTS Integration, Latency/Jitter Profiling

Simulation & CAD: Gazebo, Isaac Sim, MuJoCo, Webots, SolidWorks, CasADi/IPOPT, Ansys

EXPERIENCE

Hidroan Co.

Dec 2022 – Present

R&D Robotics Engineer — Robotics Software & Systems

Ankara, Türkiye

- Architect and sole developer of the **full autonomy stack** for a **6×6 skid-steer off-road UGV**: perception, localization, planning, tracking, and execution, in **ROS 2** and **MATLAB/Simulink**.
- Designed and implemented **GPS-denied localization**: EKF fusion of visual, LiDAR, and wheel odometry with IMU, with adaptive source weighting (wheel odometry at rest, LiDAR in motion) to hold state estimates through GPS dropout, jamming, and degraded signal.
- Developed **NMPC/MPC controllers** in **MATLAB/Simulink** (model-based) and **CasADi/IPOPT** (Python) with skid-steer kinematic and wheel-slip models; designed an active suspension controller in Simulink.
- Built the platform's **AI subsystems end-to-end**: learned perception (object detection, terrain traversability) and **RL-based control prototypes**, from data pipeline design to on-vehicle evaluation.
- Implemented **multi-UGV coordination**: leader-follower convoy navigation, formation keeping with relative pose estimation, and coordinated mission execution across multiple vehicle instances.
- Designed **collision-risk-aware navigation hooks** (overhead clearance estimation, high-rider obstacle detection) influencing route selection.
- Profiled **end-to-end navigation latency** (sensor → planner → controller → actuator) via timestamp-based tracing; optimized QoS, executor threading, and callback groups for real-time response.
- Built **Gazebo/Isaac Sim SIL environments** and executed scenario-based V&V: repeatable simulation benchmarks, log-driven failure analysis, costmap tuning, and recovery-behavior iteration.
- Owned **mechanical integration** in SolidWorks — sensor placements and mounts for the UGV sensor suite — and performed **mechanical and thermal analyses** (e.g., fin design for thermal management of onboard electronics).
- Built a **modular depth-perception unit** (Raspberry Pi + dual cameras) providing platform-independent stereo depth estimation.
- Co-authored a **TÜBİTAK 1505 university-industry project proposal** (METU–Hidroan) on autonomous navigation and sensor fusion for a semi-autonomous firefighting UGV; owner of the navigation and sensor-fusion work package.

TÜBİTAK

Aug 2022 – Feb 2023

Trainee Researcher — Robotics & ML

Ankara, Türkiye

- Contributed to a robotic mirror-therapy system: mapped human motion to robot end-effector trajectories via geometric modeling; evaluated RL-based adaptive control approaches and built experimental setups to validate convergence.

TECHNICAL PROJECTS

M.Sc. Thesis: Hierarchical DRL with Safety Filters for Off-Road UGV Navigation

2024 – Present

- Designing a **hierarchical deep RL architecture** (PPO high-level / TQC+HER low-level) with **per-wheel velocity commands** for a real 6×6 skid-steer platform; **DreamerV3-style latent world model** for sample efficiency and a **QP-based constraint-projection safety filter** enforcing actuator/stability limits.
- Advisors: Prof. Dr. E. İ. Konukseven & Asst. Prof. Ş. Ertekin Bolelli.

NMPC Energy-Optimal Path Tracking for Skid-Steer UGV

2024 – 2025

- Formulated and solved an **NMPC trajectory-tracking problem** (CasADi + IPOPT) minimizing energy consumption under kinematic constraints; demonstrated measurable **energy savings** over a Pure Pursuit baseline in field trials. **Accepted at RoManSy 2026** (presentation: Sept 2026).

Zero-Shot Terrain Traversability Analysis (DINOv2 + RELIS-3D)

2025

- Developed a **zero-shot traversability segmentation** pipeline using DINOv2 features with IncrementalPCA for memory-efficient processing on the RELIS-3D off-road dataset; strong classification without task-specific fine-tuning. **Accepted at SAVTEK** (national defense technology congress, in Turkish).

PRISM: Robust Object Detection Under Degraded Visual Conditions

2025 – Present

- Built an **adaptive image-preprocessing pipeline** (CLAHE, bilateral filtering, gamma correction) for low-light/night detection; evaluated RF-DETR on ExDark, resolving class-ID mapping bugs behind near-zero mAP. Extended into **PRISM**; manuscript under review at **The Visual Computer** (Q2).

RL Local Planner Plugin for ROS 2 Navigation

Jan 2026 – Present

- Integrated a trained RL policy as a **Nav2 local-planner module** with a lightweight **safety filter** (rule-based clamp/override); benchmarked against DWA and Pure Pursuit on success rate, path efficiency, collision count, and time-to-goal.

Interpretable World Models (CCWM) — Independent Research

2026 – Present

- Early-stage research on **causal/interpretable decomposition of world-model prediction errors**, in collaboration with Dr. Emre Akbaş (METU Computer Engineering); literature review and research proposal completed, experimental phase upcoming.

DRL-Based Vehicle Leveling / Active Suspension (PyTorch + MuJoCo / Isaac Sim)

2024 – Present

- Developing an RL-based stabilization controller for off-road platform leveling across varying terrain; targeting robust sim-to-real transfer.

Autonomous Robotic Chessboard

In progress

- Building a voice/touchscreen/app-controlled robotic chessboard: **CoreXY gantry with electromagnet actuation**, custom 3D-printed pieces, and a **three-tier motion planner** validated across thousands of simulated move sequences.

OpenRocket — Open-Source Flight Simulator Extensions

Ongoing

- Extending OpenRocket with **3D flight animation**, **6-/9-DOF flight-dynamics simulation**, flight-report generation, and improved aero modeling (upstream contribution planned).

Migrosion — ROS Package Migration Assistant

Concept / early development

- Designing a tool that tracks ROS 2 distro releases and **automatically migrates package code and configuration** to newer stable versions.

Hierarchical RL Benchmarking (HAC, SAC, TQC, DQN + HER) — Course Project

Dec 2025

- Built a reproducible evaluation protocol for sparse-reward navigation in MuJoCo: success rate, path efficiency, collision rate, energy proxy, and ablation-ready pipelines.

COMPETITIONS & LEADERSHIP

ANZU — TEKNOFEST Advanced Autonomous Systems Competition, Autonomy Lead

2026 – Present

- Leading autonomy for a coordinated **fixed-wing UAV + UGV system** in a disaster-response mission scenario: aerial perception, UAV→UGV data sharing, onboard decision-making, and autonomous task execution.
- Passed the design-report evaluation stages within a larger multidisciplinary team; **finals in 2027**.

TerraDrive — TEKNOFEST İKA (Unmanned Ground Vehicle), Team Captain

Oct 2025 – Present

- Led the 3-person team's end-to-end development of a fully autonomous **Ackermann-steered UGV** (ROS 2 Jazzy, NVIDIA Jetson Orin Nano): **classical-CV corridor-centerline perception**, waypoint generation, and Pure Pursuit tracking with adaptive lookahead — no neural networks, for real-time robustness.
- Advanced to the Critical Design Report stage of the 2026 cycle; **preparing the 2027 entry** and evaluating international UGV competitions.

NVIDIA AlpaSim E2E Closed-Loop Challenge 2026 (nuPlan track) — Individual Entry

Starting Aug 2026

- Approach planned; development starting Aug 2026 — building a closed-loop driving policy for the public leaderboard (deadline Oct 2026; NeurIPS 2026 competition track).

The BARN Challenge (ICRA) — Cluttered-Environment Navigation, Individual Entry

Planning

- Preparing an entry for the benchmark on collision-free navigation in highly cluttered environments: simulation plan defined and GitHub repository established; development ongoing.

Bernoulli FS Racing Team — Founder & Vice President

May 2023 – Jul 2024

- Led a multi-disciplinary engineering team to **4th globally** (Formula Student Romania) and **1st in Türkiye ranking**.

Yıldırım Rocket Team

Oct 2019 – Oct 2024

- 1st place** (Preliminary Design Report), 2022 national competition. CFD, structural analysis, MATLAB-based simulation.

PUBLICATIONS

Published

- **ArcNav: Real-Time Curvature-Based Map-Free Path Planning in Unstructured Environments for UGVs.** IEEE MMAR 2025. DOI: [10.1109/MMAR65820.2025.11150882](https://doi.org/10.1109/MMAR65820.2025.11150882)
- **Design and Implementation of Leveling and Balancing Algorithm.** IEEE IDITR 2024. DOI: [10.1109/IDITR62018.2024.10554321](https://doi.org/10.1109/IDITR62018.2024.10554321)
- **Dynamic Height Adjustment and Obstacle Climbing Capabilities of a 6x6 Unmanned Ground Vehicle with Advanced Hydropneumatics Suspension Systems** UMTIK 2024 (Int. Conference on Machine Design and Production).

Accepted / to appear

- **NMPC-Based Energy-Optimal Path Tracking for Skid-Steer UGV.** RoManSy 2026 (presentation: Sept 2026).
- **Zero-Shot Terrain Traversability Analysis with DINOv2.** SAVTEK 2026 (in Turkish).
- **DINOV2-BASED FEW-SHOT ANOMALY DETECTION FOR MANUFACTURING QUALITY INSPECTION: A COMPARATIVE STUDY OF FOUNDATION VISION MODELS** UMTIK 2026 (Int. Conference on Machine Design and Production).

Under review

- **Model-Optimizer Interaction in Learned-Dynamics Predictive Control: A Factorial Simulation Study on a 6×6 Unmanned Ground Vehicle** TOK 2026 (Turkish National Conference on Automatic Control).
- **How Much Can Per-Image Classical Filter Selection Help Frozen Object Detectors? A Tie-Aware Oracle, Test-Retest, and Policy-Value Analysis** The Visual Computer (Q2).
- **Image processing and artificial intelligence-based surface finish detection in conventional and ultrasonic-assisted end milling of AA7075 alloy** Proc. IMechE Part B: Journal of Engineering Manufacture (Q2).

EDUCATION

Middle East Technical University (METU)

2024 – Present

M.Sc. in Robotics (GPA: 3.57/4.00) — coursework completed, thesis stage

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- Thesis: Safe hierarchical deep RL with optimization-based safety filters for autonomous off-road UGV navigation.
- Coursework: Deep Learning, Reinforcement Learning, Machine Vision, Inertial Navigation Systems, Advanced Dynamics, Scientific ML, Design of Intelligent Machines, System Engineering, Advanced Deep Learning.

Ankara Yıldırım Beyazıt University

Jun 2024

B.Sc. in Mechanical Engineering (GPA: 3.11/4.00)

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Academic Service: Reviewer — TOK 2026 (Turkish National Conference on Automatic Control).